

samir.ghosh@ucsc.edu / samir.tech / github.com/smrghsh / 0009-0004-4575-4954

Education	University of California, Santa Cruz	Santa Cruz, California
	PhD Candidate in Computational Media · 2022 – <i>Present</i>	
	Advisor: Katherine Isbister	
	<i>Expected graduation: December 2026</i>	
	University of Southern California	Los Angeles, California
	BS in Computational Linguistics · 2018	
	BA in Cognitive Science	

Selected Projects	Virtual Reality for Scientific Sensemaking	
	Multi-year research program in collaboration with civil engineers, marine scientists, and other technical domains to explore how collaborative XR tools can augment the scientific sensemaking process. Supported by the Sloan Foundation.	
	AI War Cloud 🏆	
	Ars Electronica 2025 piece by Sarah Ciston that illustrates complex relations of AI technology across industry and government. Developed XR views using a bespoke rendering technique. Winner of the S+T+ARTS Grand Prize of the European Commission.	
	Booksnake AR	
	NEH funded iOS AR app displaying assets from the Library of Congress and other archival information. Early design, software architecture, and project management contributions.	
	Bunker Hill VR	
	Historical recreation of 1930s downtown Los Angeles using civil engineering data. Project management, documentation, and technical contributions.	
	Bodyscape	
	Internationally exhibited (e.g. SIGGRAPH, Ars Electronica) wearable technology fashion piece by Behnaz Farahi. Contributions in gait responsive algorithms, circuit design, safety systems, and generative design.	

Honors & Scholarships	Dean's Travel Grant	2025
	Finalist, UC Santa Cruz Grad Slam	2023
	Regents Fellowship	2023
	CITRIS Tech For Good Award	2022
	Dean's Fellowship	2022

Publications

AI WAR CLOUD DATABASE

Samir Ghosh and Sarah Ciston

SIGGRAPH Spatial Storytelling '26

Designing Shared Virtual Reality Sensemaking Tools for Wildfire Mitigation 🏆

DOI pending

Samir Ghosh, Maryann Godje, Charles Lesser, Yanglan Wang, Paola Lorusso, Kenichi Soga, Joshua McVeigh-Schultz, and Katherine Isbister

Meaningful XR '26

Diving in a Coral Reef: Collaborative XR Exploration of Reef Restoration using Gaussian Splats

DOI pending

Samir Ghosh, Erik Chao, Christopher Lowrie, Will Gislason, Michael Beck, Joshua McVeigh-Schultz, and Katherine Isbister

Meaningful XR '26 Demo

“Let's Jump into More Creative Avatars and Take this Brainstorm to the Flying Platform:” Playful Prototypes of VR Meeting Support Tools

Anya Osborne, Joshua McVeigh-Schultz, Alexandra Leeds, George Butler, Samir Ghosh, and Katherine Isbister

DIS 2025

Introducing Booksnake: A Scholarly App for Transforming Existing Digitized Archival Materials into Life-Size Virtual Objects for Embodied Interaction in Physical Space, using IIIF and Augmented Reality

Sean Fraga, Christy Ye, Henry Huang, Zack Sai, Michael Hughes, April Yao, and Samir Ghosh

CHI EA '25

Purposeful XR: Affordances, Challenges, and Speculations for an Ethical Future

Elizabeth Childs, Samir Ghosh, Sebastian Cmentowski, Andrea Cuadra, Rabindra Ratan

CHI EA '25

Designing Shared VR Tools for Spatial Scientific Sensemaking About Wildfire Evacuation

Samir Ghosh, Yuhui Wang, William Zhou, Kelly Lin, Joshua McVeigh-Schultz, and Katherine Isbister.

CHI EA '24

Eye Ball: Gazing as a Dilemma in a Competitive Virtual Reality Game

Michael Lankes, Samir Ghosh, Charles Bishop Lesser, and Katherine Isbister.

CHI EA '24

The Cuteness Factor: An Interpretive Framework for Artists, Designers and Engineers

Angela Y.H. Fan, Chen Ji, Ella Dagan, Samir Ghosh, Yuhui Wang, Katherine Isbister.

DIS 2023

Workshop Papers

Brahma: an Open Source Library for Cross-platform Social XR and Scientific Visualization

Samir Ghosh, Joshua McVeigh-Schultz, and Katherine Isbister

Social Augmentation through XR Technologies Workshop, CHI '26

Designing Interaction Approaches for Shared Sensemaking in XR

Samir Ghosh, Joshua McVeigh-Schultz, and Katherine Isbister.

Sensemaking Workshop, CHI '24

Social Physiological Data Awareness in Collocated Mixed Reality Movement

Samir Ghosh, Charles Lesser, Kaia Rae Schweig, Sofia De La Vega Mireles, and Katherine Isbister.

PhysioCHI Workshop, CHI '24

Revealing Aspects of Hawai'i Tourism Using Situated Augmented Reality

Karen Abe, Jules Park, and Samir Ghosh

With or Without Permission Workshop, CHI '24

Designing a mixed-initiative multi-user VR interface for wildfire mitigation

Samir Ghosh, Yanglan Wang, Kecheng Cheng, Anthony Angeles, Andrew Moskovich, Kenichi Soga, and Katherine Isbister.

HCI for Climate Change Workshop, CHI 2023

Research Positions	Graduate Student Researcher, UC Santa Cruz Coastal Climate Resilience Center and UC Santa Cruz Visualization Lab	Jun 2025 – Present
	Graduate Student Researcher, UC Santa Cruz Catalyst Program, UC Open Source Program Office	Jun 2025 – Present
	Visiting Scholar, Blekinge Institute of Technology HINTS Project, Department of Computer Science	May 2025 – Jun 2025
	Graduate Student Researcher, UC Santa Cruz Virtual Reality for Scientific Sensemaking, Social Emotional Technology Lab Supported by the Sloan Foundation	Jun 2023 – Apr 2025
Industry Experience	Ahmanson Lab USC Harman Academy Assistant Director · Jan 2019 – Aug 2022	Los Angeles, California
	YUR Inc. VR Developer · Jul 2021 – Dec 2021	Los Angeles, California
	Intel Corporation Intern · Summer 2016, Summer 2018, Fall 2018	Santa Clara, California
	Enlighted Inc. Intern · Summer 2014	Sunnyvale, California
Teaching	Teaching Assistant, UC Santa Cruz CMPM 115: Lead By Design	Winter 2022

Seminars, Talks & Tutorials	Seals, Proteins, and Wildfires <i>Lightning Talk, Questioning Reality '26 (QR'26)</i>	May 2026	3D Modeling Basics <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Sep 2021
	A Career in VR + Coding and Art Workshop <i>Bullis Charter School, Los Altos, California</i>	Jan 2026	Glitch + D3.js <i>Generative art-a-thons, Ahmanson Lab at USC</i>	Oct 2020
	Virtual Reality and Scientific Sensemaking <i>Department of Industrial Design, TU Eindhoven, Netherlands</i>	Jun 2025	VR with Mozilla Hubs <i>Generative art-a-thons, Ahmanson Lab at USC</i>	Oct 2020
	Virtual Reality and Scientific Sensemaking <i>German Aerospace Agency (DLR), Braunschweig, Germany</i>	Jun 2025	p5js <i>Generative art-a-thons, Ahmanson Lab at USC</i>	Sep 2020
	Virtual Reality and Scientific Sensemaking <i>Department of Computer Science, Blekinge Institute of Technology, Sweden</i>	May 2025	Applied Neural Networks <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Apr 2020
	Revisiting the for Loop <i>Slugworks, UC Santa Cruz</i>	Nov 2023	Introduction to Creative Code <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Apr 2020
	A Career in HCI and VR <i>Cognitive Science Student Association, UC Santa Cruz</i>	Oct 2023	STEM Speaker Series <i>Katherine Johnson STEM Academy</i>	Mar 2020
	Generative Art in Virtual Reality Using p5js <i>Digital Arts and New Media, UC Santa Cruz</i>	Jun 2023	Deepfake Detection <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Mar 2020
	Wildfires in Virtual Reality <i>UC Santa Cruz Grad Slam, Kuumbwa Jazz Center</i>	Mar 2023	Practical Arduino <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Mar 2020
	Multi-user VR Workshop <i>Digital Arts and New Media, UC Santa Cruz</i>	Feb 2023	Data Surveillance and Digital Rights <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Feb 2020
	Surveillance and the Attention Economy <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Mar 2022	Introduction to 3D Printing and the Makerbot Replicator <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Feb 2020
	Computational Art <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Feb 2022	WebVR <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Nov 2019
	WebRTC, WebGL, and other web protocols <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Jan 2022	Wearable Technology <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Oct 2019
	Techniques with Graphics Code <i>Emergent Technology Series, Ahmanson Lab at USC</i>	Sep – Oct 2021	Get Your Own Climate Data <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Oct 2019
	VR Web Development <i>Emergent Technology Series, Ahmanson Lab at USC</i>	Sep – Oct 2021	Introduction to 3D Printing and the Makerbot Replicator <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Sep 2019
	Object Recognition, Privacy Rights, and Data Collection <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Sep 2021	Practical Arduino <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Sep 2019
	Sensors, Lights, and Motors <i>Polymathic Making Workshops, Ahmanson Lab at USC</i>	Sep 2021	Promise and Peril of Algorithmic Living <i>USC Visions and Voices</i>	Apr 2018

Service	Google Summer of Code Mentor · <i>UC Open Source Program Office</i>	Summer 2025
	Creative Code Collective Community Lead	2023 – Present
	Committee for Planning and Budget Graduate Representative · <i>Academic Senate, UC Santa Cruz</i>	2023 – Present
	MIT Reality Hack Mentor	2025
	ACM Halfway to the Future Symposium 2024 Social Media Chair	2023 – 2024
	Google Summer of Code Mentee, Contributor · <i>Processing Foundation</i>	Summer 2022
	Corpus Callosum (Student Org.) Technical Director · <i>Viterbi School of Engineering, USC</i>	Fall 2015 – Spring 2018

**Personal
Information**

Citizenship	USA
Languages	English (native), French (proficient), Korean (basic)
Email	samir.ghosh@ucsc.edu
Website / Portfolio	samir.tech
Interests	Dance, yoga, creative code